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Assignment -05
Image Processing and Pattern
Recognition
TCS-361

Q1.) A grayscale image has the following pixel distribution:

Gray levels: 0, 1, 2, 3

Number of pixels: 5, 10, 15, 20

Perform histogram equalization and calculate the new intensity values.

Ans.) Given:

Gray levels: 0, 1, 2, 3

Pixels: 5, 10, 15, 20

Total $N = 50$, $L = 4$

Step 1: Probability

$$p(0) = 5/50 = 0.1$$

$$p(1) = 10/50 = 0.2$$

$$p(2) = 15/50 = 0.3$$

$$p(3) = 20/50 = 0.4$$

Step 2: CDF

$$CDF(0) = 0.1$$

$$CDF(1) = 0.3$$

$$CDF(2) = 0.6$$

$$CDF(3) = 1.0$$

Step 3: New values

$$s_k = (L-1) * CDF = 3 * CDF$$

$$s(0) = 3 \times 0.1 = 0.3 = 0$$

$$s(1) = 3 \times 0.3 = 0.9 = 1$$

$$s(2) = 3 \times 0.6 = 1.8 = 2$$

$$s(3) = 3 \times 1 = 3$$

New intensity levels:

$$0 \rightarrow 0$$

$$1 \rightarrow 1$$

$$2 \rightarrow 2$$

$$3 \rightarrow$$

Q2.) Discuss the real-world applications of digital image processing in medical imaging, satellite imaging, and security systems.

Ans.)

Medical Imaging:

- MRI, CT scan analysis
- Tumor detection
- Image enhancement for diagnosis

Satellite Imaging:

- Weather forecasting
- Land use analysis
- Disaster monitoring

Security Systems:

- Face recognition
- Fingerprint identification
- Surveillance and motion

detection

Q3.) Explain the CMY color model and how it is used in digital images.

Ans.)

CMY stands for Cyan, Magenta, Yellow.

It is a subtractive color model used in printing.

Relation with RGB:

$$C = 1 - R$$

$$M = 1 - G$$

$$Y = 1 - B$$

(or in 8-bit images:

$$C = 255 - R, M = 255 - G, Y = 255 - B)$$

Used in:

- Printers
- Ink-based color reproduction